Exam 4 Raina Swamy

64

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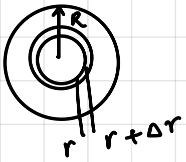



(i) Assumptions: Spherical and Isothermal and $C_A(r=R) = C_{AS}$

Concentration profile : $C_A(r) = \frac{-k}{6 D_{AB} r^2} - \frac{A}{D_{AB} r} + C_2$

(7/20)

B.C.1 $\Rightarrow$ $r=0$ $\rightarrow$ $C_A : 0$

$C_A$ is FINITE does NOT imply $C_A$ is 0.

B.C.2 $\Rightarrow$ $r=R$ $\rightarrow$ $C_A = C_{AS}$

$$-\frac{C_1}{D_{AB} r} = 0$$

For B.C.1 $C_1 = 0$

$$C_{A0} = \frac{-K}{6 D_{AB}} R^2 + C_2$$

For B.C.2

$$C_2 = \frac{K}{6 D_{AB}} K^2 + C_{A0}$$

$$\boxed{C_A(r) = \frac{-K}{6 D_{AB} r^2} - \frac{A}{D_{AB} r} + C_{A0}}$$ x

(iv) The condition at which diffusion is the overall rate limiting process would be if

(2/10)

$K_1'' \rightarrow \infty$

$$N''_{A2=0} \approx \frac{2 C D_{AA_2}}{\delta} \ln\left(\frac{1}{1 - x_{A0}/2}\right)$$ x

(iii) (1/10) If $K''_1 \rightarrow 0$ there would be a reaction controlled process so it can be used for kinetic studies x

$\sqrt{\frac{k_s}{D C_{As}}} R \ll 1$

(ii)

$$\frac{1}{r^2}\frac{d}{dr}\left[r^2 \frac{dc}{dr}\right] = \phi^2 R_1 (c, T)$$

(8/20)

Hence if T remained constant

$$\frac{1}{r^2}\frac{d}{dr}\left[r^2 \frac{dc}{dr}\right] = \phi^2 R_1 [c]$$ x

②

$T_i = 27^\circ C$

$\dot{m} = 450 \ kg/h$

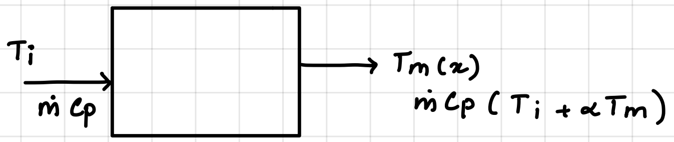
$q'(W/m) = \alpha \cdot x$

$\alpha = 20 \ W/m^2$

$x(m) \rightarrow$ axial distance from the tube entrance

$C_p = 4.18 \ kJ/kg \cdot K$

$x = 30 m$




Temperature distribution for $T_m(x)$

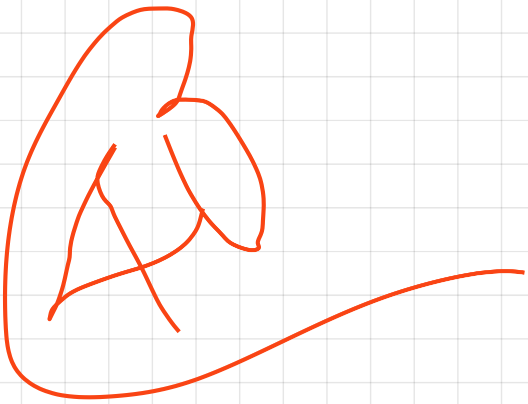
$$q = \dot{m} \, C_p \, d \, T_m$$

$$q = q' \cdot dx$$

$$\alpha x \cdot dx = \dot{m} \, C_p \, dT_m$$

integrating

$$\alpha \int_0^x x \, dx = \dot{m} \, C_p \int_{T_{m,i}}^{T_m} dT_m$$



we get,

$$\boxed{T_m(x) = T_i + \frac{\alpha x^2}{2 \dot{m} C_p}}$$

Outlet temperature

$$T_{m,o} = T_{mi} + \frac{a x^2}{2 \dot{m} C_p}$$

$$= (27 + 273.15) K + \frac{(20)(30)^2 (3600 s)}{2 \times 450 \times (4.179 \times 10^3)}$$

$$= 300.15 K + \frac{64800000}{3761100}$$

$$= 300.15 K + 17.23$$

$$\boxed{\begin{aligned} &= 317.38 \ K \\ &= 44.23^\circ C \end{aligned}}$$